

ASSIGNEMENT II
ADVANCED PROGRAMMING
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ROLLNO-CSB24045

Write a c program for three different operations as stated in question 1 to analyze the space complexity

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    printf("Enter the size of input (n): ");
```

```
    scanf("%d", &n);
```

```
    printf("\n---- Space Complexity Demonstration ----\n");
```

```
    printf("\n1. Constant Space O(1):\n");
```

```
    int a = 10;
```

```
    int b = 20;
```

```
    int sum = a + b;
```

```
printf("Sum of two numbers = %d\n", sum);  
printf("Memory used does NOT depend on n.\n");
```

```
printf("\n2. Linear Space O(n):\n");
```

```
int *arr;  
arr = (int*) malloc(n * sizeof(int));
```

```
if(arr == NULL)  
{  
    printf("Memory allocation failed!\n");  
    return 1;  
}
```

```
for(int i = 0; i < n; i++)  
{  
    arr[i] = i + 1;  
}
```

```
printf("Array of %d elements created.\n", n);  
printf("Memory used increases linearly with n.\n");
```

```
free(arr);
```

```
printf("\n3. Quadratic Space  $O(n^2)$ :\n");
```

```
int **matrix;
```

```
matrix = (int**) malloc(n * sizeof(int*));
```

```
for(int i = 0; i < n; i++)  
{  
    matrix[i] = (int*) malloc(n * sizeof(int));  
}
```

```
for(int i = 0; i < n; i++)  
{  
    for(int j = 0; j < n; j++)  
    {  
        matrix[i][j] = i + j;  
    }  
}
```

```
printf("Matrix of size %d x %d created.\n", n, n);
```

```
printf("Memory used increases quadratically with n.\n");
```

```
for(int i = 0; i < n; i++)  
{  
    free(matrix[i]);  
}
```

```
}  
free(matrix);  
  
printf("\nProgram Finished Successfully!\n");  
  
return 0;  
}
```

OUTPUT

Enter the size of input (n): 5

---- Space Complexity Demonstration ----

1. Constant Space $O(1)$:

Sum of two numbers = 30

Memory used does NOT depend on n.

2. Linear Space $O(n)$:

Array of 5 elements created.

Memory used increases linearly with n.

3. Quadratic Space $O(n^2)$:

Matrix of size 5 x 5 created.

Memory used increases quadratically with n.

Program Finished Successfully!